

CN7050 – Intelligent Systems

Week 9 Lab – Recommender Systems using ALS

This lab is worth 5 marks. It focuses on implementing a movie recommender system using the Alternating Least Squares (ALS) algorithm in PySpark.

Task 1 (2 marks): Hyperparameter Tuning

Train an ALS model with different combinations of parameters such as rank and regParam. Select the best model based on the lowest RMSE (Root Mean Squared Error).

Task 2 (2 marks): Generate Top-10 Recommendations

Using the best model from Task 1, generate Top-10 movie recommendations for a given user (e.g., userId = 123).

Task 3 (1 mark): Analyse Recommended Genres

Join recommendations with the movie metadata to identify which genres are most frequently recommended to the user.

The code does not have to match exactly with the template provided in the lab materials, but your methodology should follow the same principles. Use the best parameters identified in Task 1 for completing Task 2 and Task 3.